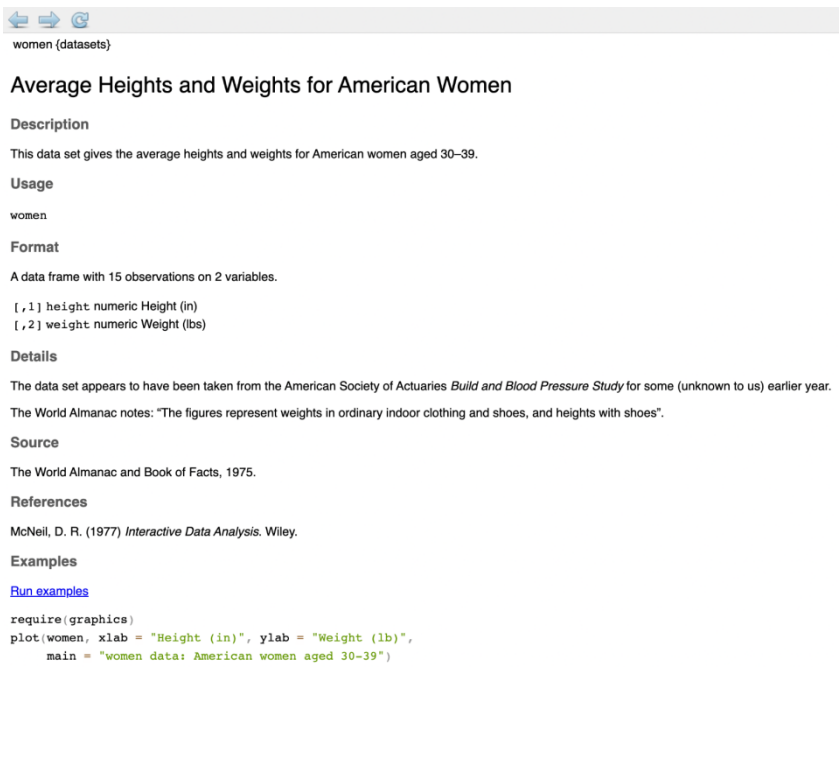
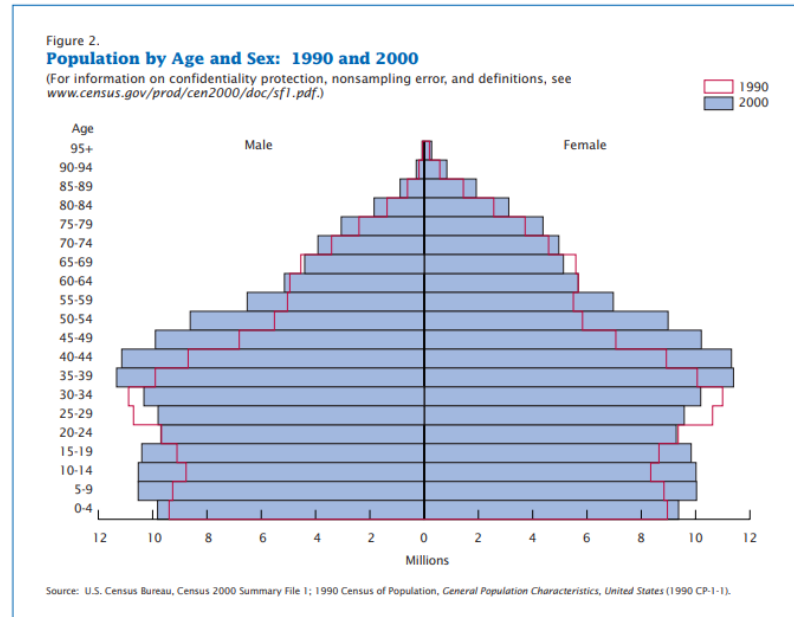


Name:	Zera 
Date:	07.25.2023
Instructor:	Samuel Chukwuemeka
Project:	Inferential Statistics
1st Dataset: women	1st Source: R/R Studio Dataset: women Description: This data set gives the average heights and weights for American women aged 30-39. The data set appears to have been taken from the American Society of Actuaries <i>Build and Blood Pressure Study</i> for some (unknown to us) earlier year. The World Almanac and Book of Facts, 197 <pre> > women height weight 1 58 115 2 59 117 3 60 120 4 61 123 5 62 126 6 63 129 7 64 132 8 65 135 9 66 139 10 67 142 11 68 146 12 69 150 13 70 154 14 71 159 15 72 164 > </pre>

	 women (datasets) Average Heights and Weights for American Women Description This data set gives the average heights and weights for American women aged 30–39. Usage <pre>women</pre> Format A data frame with 15 observations on 2 variables. <pre>[,1] height: numeric Height (in) [,2] weight: numeric Weight (lbs)</pre> Details The data set appears to have been taken from the American Society of Actuaries <i>Build and Blood Pressure Study</i> for some (unknown to us) earlier year. The World Almanac notes: "The figures represent weights in ordinary indoor clothing and shoes, and heights with shoes". Source The World Almanac and Book of Facts, 1975. References McNeil, D. R. (1977) <i>Interactive Data Analysis</i>. Wiley. Examples Run examples <pre>require(graphics) plot(women, xlab = "Height (in)", ylab = "Weight (lb)", main = "women data: American women aged 30-39")</pre>
Objectives: women	(1.) Construct a 95% confidence interval for the population mean of the weights of American women aged 30 – 39 in the year 2000. (2.) Interpret the confidence interval. (3.) Estimate the population mean of the weights of American women aged 30 – 39 in the year 2000.
1st Parameter:	(a.) Estimating Population Mean (1.) Variable: Weight Unit of variable: pounds Sample Size: 15 Sample: 15 American women aged 30-39 in year 2000 (2.) Population: All American women aged 30-39 in year 2000 Assume Year: 2000 (Unknown year, this is assumption year) (3.) Parameter to estimate: Population mean Test: <i>t-test</i> (4/5.) Verify Requirements for Test:

1. The sample is a simple random sample.
2. The population is normally distributed.
3. The sample size is less than 5% of the population size.



2

U.S. Census Bureau

(6.) Reason for test: The population standard deviation was not given.

(7.)

```
> t.test(women$weight)
```

One Sample t-test

```
data:  women$weight
t = 34.168, df = 14, p-value =
6.907e-15
alternative hypothesis: true mean is not
equal to 0
95 percent confidence interval:
 128.1504 145.3162
sample estimates:
mean of x
 136.7333
```

(8.) Run a t-test (`t.test ( )` function) with a 95% confidence interval for population mean from the sample data.

Since we are only concerned about the weight of the women, we will code: `t.test (women$weight)`

(9.) Based on the results:

The test statistic: 34.168

The degrees of freedom: 14

The sample mean: 136.7333

The 95% confidence interval: 128.1504, 145.3162

We are 95% confident the population mean of the weights of American women aged 30-39 in the year 2000 is between 128.1504 pounds and 145.3162 pounds.

(10.) This implies that in about 95% of all the samples of American women aged 30-39 in the year 2000, the confidence interval will contain the population mean of (128.1504, 145.3162) pounds.

2nd Dataset: quakes	2nd Source: R/R Studio Dataset: quakes Description: The data set gives the locations of 1000 seismic events of MB > 4.0. The events occurred in a cube near Fiji since 1964. The data frame with 1000 observations on 5 variables. [,1] lat numeric Latitude of event [,2] long numeric Longitude [,3] depth numeric Depth (km) [,4] mag numeric Richter Magnitude [,5] stations numeric Number of stations reporting There are two clear planes of seismic activity. One major plate junction; the other is the Tonga Trench off New Zealand. These data constitute a subsample from a larger dataset of containing 5000 observations. This is one of the Harvard PRIM-H project data sets. They in turn obtained it from Dr. John Woodhouse, Dept. of Geophysics, Harvard University.
--------------------------------	---

```
Console Terminal Background Jobs
R 4.3.1 · ~/Project Dataset (quakes) (zera williams)/

> quakes

      lat  long depth mag
1  -20.42 181.62   562 4.8
2  -20.62 181.03   650 4.2
3  -26.00 184.10    42 5.4
4  -17.97 181.66   626 4.1
5  -20.42 181.96   649 4.0
6  -19.68 184.31   195 4.0
7  -11.70 166.10    82 4.8
8  -28.11 181.93   194 4.4
9  -28.74 181.74   211 4.7
10 -17.47 179.59   622 4.3
11 -21.44 180.69   583 4.4
12 -12.26 167.00   249 4.6
13 -18.54 182.11   554 4.4
14 -21.00 181.66   600 4.4
15 -20.70 169.92   139 6.1
16 -15.94 184.95   306 4.3
17 -13.64 165.96    50 6.0
18 -17.83 181.50   590 4.5
19 -23.50 179.78   570 4.4
20 -22.63 180.31   598 4.4
```

these stats continue 1-1000 earthquakes

quakes (datasets) R Documentation

Locations of Earthquakes off Fiji

Description

The data set give the locations of 1000 seismic events of MB > 4.0. The events occurred in a cube near Fiji since 1964.

Usage

quakes

Format

A data frame with 1000 observations on 5 variables.

[,1] lat numeric Latitude of event

[,2] long numeric Longitude

[,3] depth numeric Depth (km)

[,4] mag numeric Richter Magnitude

[,5] stations numeric Number of stations reporting

Details

There are two clear planes of seismic activity. One is a major plate junction; the other is the Tonga trench off New Zealand. These data constitute a subsample from a larger dataset of containing 5000 observations.

Source

This is one of the Harvard PRIM-H project data sets. They in turn obtained it from Dr. John Woodhouse, Dept. of Geophysics, Harvard University.

Examples

Objectives: quakes	(1.) Construct a 95% confidence interval for the population proportion of the seismic events with a magnitude of 4.5ergs and greater in the year 1964. (2.) Interpret the confidence interval. (3.) Estimate the population proportion of the seismic events with a magnitude of 4.5ergs and greater in the year 1964.
2nd Parameter:	(b.) Estimating Population Proportion (1.) Variable: Richter Magnitude Unit of variable: Energy (ergs) Sample size: 1000 Sample: 1000 seismic events of MB > 4.0 that occurred in a cube near Fiji since 1964. (2.) Population: All seismic events of MB > 4.0 that occurred in 1964. (3.) Parameter to estimate: Population proportion Test: <i>prop-test</i> (4/5.) Verify Requirements for Test: 1. The sample is a simple random sample. 2. The population is normally distributed. (6.) Reason for test: The population standard deviation was not given. (7.)

```
Console Terminal Background Jobs
R 4.3.1 · ~/Project Dataset (quakes) (zera williams)/

> #To find the proportion of this population with quakes with
a magnitude of 4.5 ergs and greater
> #First we need to look at quakes$mag
> quakes$mag
 [1] 4.8 4.2 5.4 4.1 4.0 4.0 4.8
 [8] 4.4 4.7 4.3 4.4 4.6 4.4 4.4
[15] 6.1 4.3 6.0 4.5 4.4 4.4 4.5
[22] 4.2 4.4 4.7 5.4 4.0 4.6 5.2
[29] 4.5 4.4 4.6 4.7 4.8 4.0 4.5
[36] 4.3 4.5 4.6 4.1 4.4 4.7 4.6
[43] 4.4 4.3 4.6 4.9 4.5 4.4 4.3
[50] 5.1 4.2 4.0 4.6 4.3 4.2 4.4
[57] 4.5 4.0 4.4 4.3 4.7 4.1 5.0
[64] 4.6 4.9 4.7 4.1 5.0 4.5 5.5
[71] 4.0 4.5 4.3 5.2 4.4 4.3 4.1
[78] 4.5 4.2 5.3 5.2 4.5 4.6 4.3
[85] 4.0 4.3 4.7 4.5 4.2 4.3 5.1
[92] 4.7 5.2 4.2 4.2 4.0 4.5 5.2
[99] 5.1 4.7 4.1 4.6 4.7 4.7 4.6
[106] 4.2 4.4 4.6 5.7 5.0 4.5 4.2
[113] 4.0 4.8 4.4 4.2 5.3 4.7 4.8
[120] 4.2 4.8 4.3 4.7 4.5 4.4 5.1
```



```

Console Terminal Background Jobs
R 4.3.1 ~ /Project Dataset (quakes) (zera williams)
> #In order to sort this in ascending order we need to use the
sort function.

```

```

> sort(quakes$mag)
[1] 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0
[14] 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0
[27] 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.0
[40] 4.0 4.0 4.0 4.0 4.0 4.0 4.0 4.1 4.1 4.1 4.1 4.1 4.1
[53] 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1
[66] 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1
[79] 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1
[92] 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.1 4.2 4.2 4.2
[105] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2
[118] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2
[131] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2
[144] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2
[157] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2
[170] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2
[183] 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4.3 4.3 4.3
[196] 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3
[209] 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3
[222] 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3
[235] 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3 4.3

```

```

Console Terminal Background Jobs
R 4.3.1 ~ /Project Dataset (quakes) (zera williams)
[896] 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2
[911] 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.3 5.3
[924] 5.3 5.3 5.3 5.3 5.3 5.3 5.3 5.3 5.3 5.3 5.3 5.3 5.3
[937] 5.3 5.3 5.3 5.3 5.3 5.3 5.4 5.4 5.4 5.4 5.4 5.4 5.4
[950] 5.4 5.4 5.4 5.4 5.4 5.4 5.4 5.4 5.4 5.4 5.4 5.4 5.4
[963] 5.5 5.5 5.5 5.5 5.5 5.5 5.5 5.5 5.5 5.5 5.5 5.5 5.5
[976] 5.5 5.6 5.6 5.6 5.6 5.6 5.6 5.6 5.6 5.6 5.6 5.7 5.7
[989] 5.7 5.7 5.7 5.7 5.7 5.9 5.9 6.0 6.0 6.0 6.1 6.4

> #Next this becomes our numerator
> numerator<-sort(quakes$mag)
> #4.5 magnitude starts at the 378th sample till the 1000th, we want
only these samples to become our new numerator
> newnumerator<-numerator[378:1000]
> newnumerator
[1] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[14] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[27] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[40] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[53] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[66] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[79] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5
[92] 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5

```

```

Console Terminal Background Jobs
R 4.3.1 ~ /Project Dataset (quakes) (zera williams)/
[612] 5.7 5.7 5.7 5.7 5.7 5.9 5.9 6.0 6.0 6.0 6.1 6.4
> #We can not sum the numbers remaining because they are each their own
measurement. However there is 623 samples in our newnumerator. Meaning there
is 623 quakes with a magnitude with a 4.5ergs or greater.
> #Now we have all the variables needed for our prop-test. (x) = 623, (n)= 1
000 and because our sample size is greater than 30, so we will code correct =
FALSE in our test.
> prop.test(623,n=1000,correct=FALSE)

1-sample proportions test without continuity
correction

data: 623 out of 1000, null probability 0.5
X-squared = 60.516, df = 1, p-value =
7.298e-15
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.5925457 0.6525129
sample estimates:
      p
0.623

```

(8.)**Italics are used to show what codes are written into R/R Studio**

To run a prop-test, we need the following variables:

$$\hat{p} = \frac{x}{n}$$

$\hat{p}$ = sample proportion or estimated proportion of successes

x = number of individuals in the sample with specified characteristic

n = sample size

To find the proportion of this population that have quakes with a magnitude of 4.5 ergs and above. First, we need to only look at *quakes\$mag*

Then we need to sort them into ascending order using the function *sort*.

sort(quakes\$mag).

This becomes our numerator.

numerator <- sort(quakes\$mag).

	The recorded quakes with a 4.5 magnitude and above start at the 378th sample and continue till the 100th. We only want these in our numerator. <pre>newnumerator<- numerator[378:1000]</pre> We cannot sum the numbers remaining because they are each their own measurement. However, in our newnumerator there is 623 samples. Now we have all the variables needed to run our prop-test. <pre>x = 623 n = 1000</pre> Since our sample size is greater than 30, we will code correct=FALSE. <pre>prop.test(623,n=1000,correct=FALSE)</pre> (9.) Based on the results: The sample proportion (point estimate) is 0.623 The 95% confidence interval is (0.5925457, 0.6525129) We are 95% confident the population percentage of the seismic events with a magnitude of 4.5ergs and greater in the year 1964 is between 59.25457% and 65.25129%. (10.) This implies that in about 95% of all the samples of seismic events with a magnitude of 4.5ergs and greater in the year 1964, the confidence interval will contain the population percentage of (59.25457%, 65.25129%).
References: APA	Chukwuemeka, Samuel Dominic (2023). <i>R and RStudio Statistics Software</i>. Retrieved from https://statistical-science.appspot.com/ Triola, M. F. (2022). <i>Elementary Statistics</i>. (14th ed.) Hoboken: Pearson.